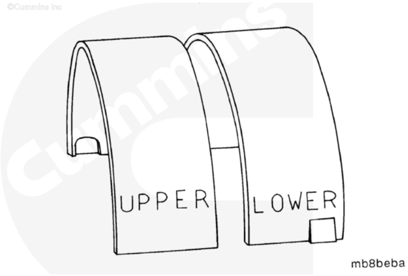
Trainings

Install

⚠ CAUTION ⚠

To reduce the possibility of engine damage, the upper and lower bearings must be installed in the correct location. The upper bearing has an oil groove. The bearing shells are marked with the words "upper" and "lower" for identification.



⚠ CAUTION ⚠

To correctly position the bearing and prevent engine damage, the bearing tang (1) must be in the slot (2) of the bearing saddle.

⚠ CAUTION ⚠

Only use Loctite™ 518 anaerobic sealant on the main cap joint mating surface. Other sealants can become hard, brittle, and allow oil and debris into the main bearing/block joint.

⚠ CAUTION ⚠

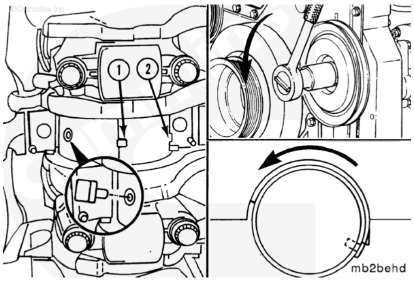
The bead must be 3 to 5 mm [0.12 to 0.2 in] wide and must not enter the main bearing shell inside diameter. Sealant in the main bearing can cause engine damage.

Used bearings **must** be installed in the same location from which they were removed.

Use Lubriplate™ 105 multi-purpose lubricant, or equivalent, to coat the inside diameter of the upper main bearing shell.

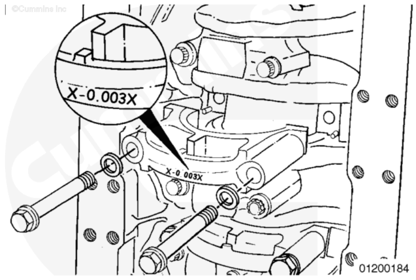
Install the upper main bearing shells (2) using the same method used to remove them.

The tang side (1) of the bearing **must** go into place last.



⚠ CAUTION ⚠

Low oil pressure faults can occur if more than one set undersize main bearings are installed in the same engine.



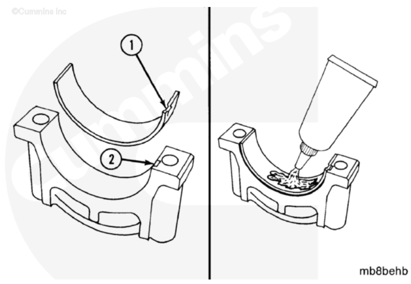
Note : Both upper and lower undersize main bearing shells must be installed as a set in the same block location.

Undersize main bearing installation procedures are the same as standard main bearings procedures. Both upper and lower undersized main bearing shells **must** be installed as a set in the same block location.

For undersize main bearing installation, mark the outside back of the main cap X -0.003 X with a permanent white Dykem™ marker to identify bearing location for future repairs. Allow the Dykem™ to dry before oil is added.

Install the lower main bearing shells with the bearing tang (1) in the slot (2) of the main bearing cap.

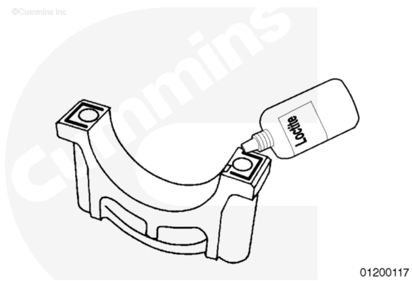
Use Lubriplate™ 105 multi-purpose lubricant, or equivalent, to coat the inside diameter of the bearing shells.



Apply Loctite® 518 anaerobic sealant to four locations on each main bearing cap.

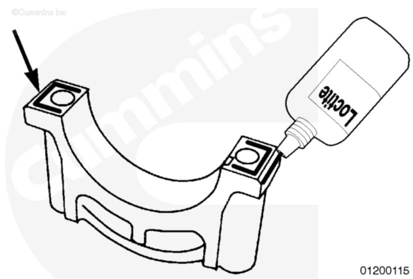
Apply a bead of Loctite® 518 anaerobic sealant to the main bearing cap mounting surface, as shown.

Repeat this procedure on the opposite side of the main bearing cap.



⚠ CAUTION ⚠

The bead must be 3 to 5 mm [0.12 to 0.2 in] wide and must not enter the main bearing shell inside diameter. Sealant in the main bearing can cause engine damage.

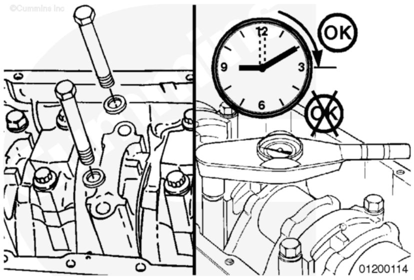


Locate the leading edge of the main bearing cap clearance chamfer.

Apply a bead of Loctite® 518 anaerobic sealant to the leading edge of the main bearing cap clearance chamfer (as shown).

Repeat this procedure on the opposite side of the main bearing cap.

To reduce the possibility of bore size issues, clearance issues, or both, the main bearing capscrews **must** be tightened within 15 minutes of the Loctite® 518 anaerobic sealant application.

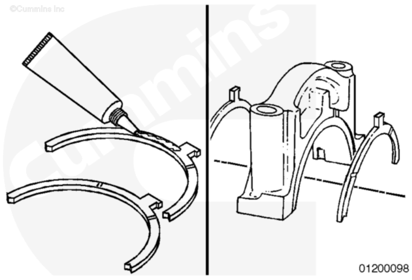


Use Lubriplate™ 105 multi-purpose lubricant, or equivalent, to coat the lower thrust bearings.

Install the lower thrust bearings in the number 4 main bearing cap, as shown.

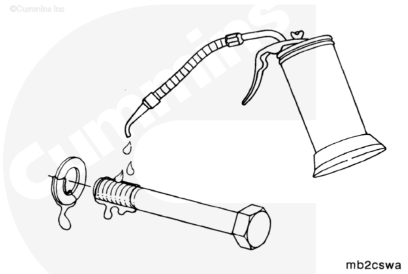
The grooves of the thrust bearing **must** be toward the crankshaft.

The locating dowels **must not** protrude above the thrust bearing surface.

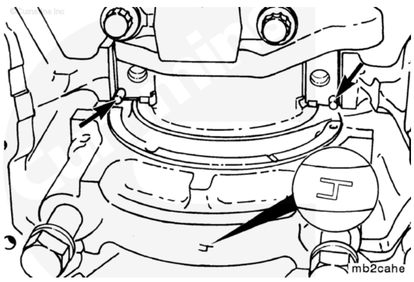


Use clean 15W-40 oil to coat the capscrew threads and both sides of the washers.

Drain the excess oil from the capscrews before installing them in the cylinder block.



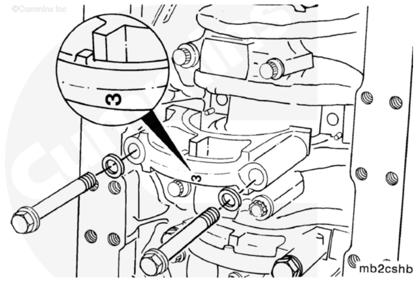
The number 4 main bearing cap **must** be aligned with the dowel pins in the bearing saddle when the capscrews are tightened.



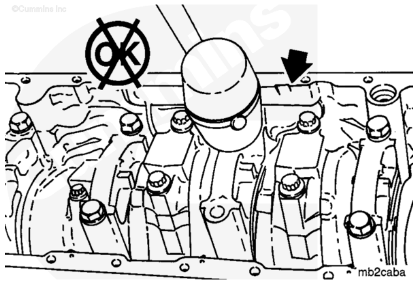
The main bearing caps are numbered 1 through 7 from front to rear in the cylinder block. The caps **must** be installed so the number on the cap matches the bearing saddle in the block.

The lock tangs in the main bearing saddle and bearing cap **must** be on the same side.

Install the main bearing caps.

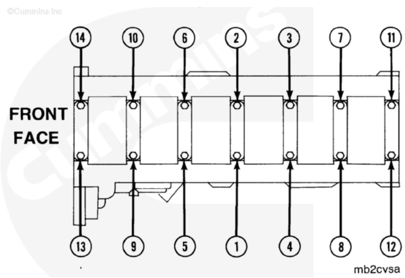


Do **not** hit the main bearing cap with a hammer.



⚠ CAUTION ⚠

When using new main bearing caps with older M11 and ISM cylinder blocks, only use the torque-plus-angle method when tightening the main bearing capscrews. The torque-to-yield procedure will damage cylinder blocks stamped 00622080 and lower.



Torque Plus Angle procedure (Engine Block Numbers Lower Than 00622081

Tighten the main bearing capscrews in alternating sequence to the following torque values:

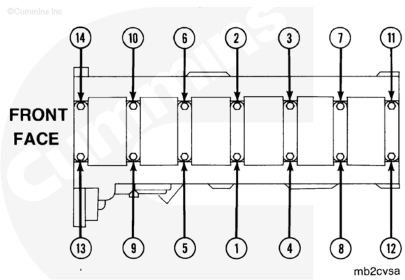
Torque Value:

1. 68 n•m [50 ft-lb]
2. 142 n•m [105 ft-lb]
3. 210 n•m [155 ft-lb]
4. Loosen completely
5. Repeat steps 1 through 3.

Torque-to-Yield Procedure

Note : Only use the torque-to-yield procedure to tighten the main bearing cap capscrews on the following engines:

- Engine serial numbers 35011095 and higher.
- Engine cylinder blocks stamped 00622081 and higher.



- Blocks with TTY stamped on the rear engine serial number stamp pad.

Tighten the main bearing cap capscrews with the following steps:

Torque Value:

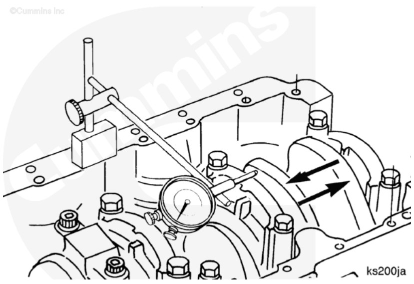
1. 95 n•m [70 ft-lb]
2. 135 n•m [100 ft-lb]
3. Loosen completely
4. 95 n•m [70 ft-lb]
5. 135 n•m [100 ft-lb]
6. Rotate 180 degrees.

To rotate 180 degrees, use a socket that has been etched with two axial lines 180 degrees apart.

Place the socket over the main cap capscrew head, and mark the cap with a lead pencil at the location of the socket axial line number 1.

Use an impact wrench to rotate the main bearing capscrew in the direction of tightening until the axial line number 2 lines up with the mark on the cap (180 degrees).

Use a dial indicator to measure the crankshaft end clearance.



Crankshaft End Clearance		
mm		in
0.10	MIN	0.004
0.56	MAX	0.022

If the end clearance is **not** within the specifications, inspect the main bearing cap and thrust bearing surfaces in the cylinder block.

Last Modified: 09-Mar-2015